Diversity-Guided Distillation With Modality-Center Regularization for Robust Multimodal Remote Sensing Image Classification

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***Abstract*— Multimodal learning has shown great potential in remote sensing image classification and attracted increasing interest in the community. Although it is preferable to collect multiple modalities for training, not all of them are available in practical scenarios. To this end, we propose a general diversity-guided distillation network (DGDNet) with modality-center regularization (MCR) to facilitate accurate model inference when modalities are missing. Compared with existing modality reconstruction methods, DGDNet does not need prior knowledge of the missing modality and can handle various missing modalities via only one model. Specifically, DGDNet consists of two components: the deployment network extracting the modality-invariant representation for robust inference and the teacher network transferring comprehensive multimodal information to the deployment network. This enables the deployment network to learn the modality invariant and specific information simultaneously while maintaining robustness for incomplete modality input. In particular, we design a novel diversity-guided distillation (DGD) method that transfers knowledge by matching the feature diversity. This helps overcome the representation heterogeneity when encouraging the deployment network to learn modality-specific information. Besides, an MCR strategy is proposed to address the unbalanced training of teacher and deployment networks by constraining the intra-class inter-modality variations. This helps alleviate the underfitting for weak modality, improving the model performance. Finally, extensive experiments demonstrate that the proposed DGDNet can address the problem of missing modalities effectively and achieves state-of-the-art performance. The code is available at https://github.com/shicaiwei123/TGRS-DGDNet.**

***Index Terms*— Classification, deep learning, distillation knowledge, missing modalities, multimodal, remote sensing.**

I. Introduction

REMOTE sensing image classification is crucial for various applications in Earth observation, such as urban planning [1], land use analysis [2], and environmental monitoring [3]. Propelled by the advancements in earth observation technologies, classification models that leverage multiple modalities show great power in practice and gain growing attention from the research community [4], [5], [6], [7], [8], [9], [10]. However, most successful methods assume that the models are trained and tested with the same modality data. This is not always possible in real-world deployment scenarios due to sensor failure [11], data corruption [12], and mismatched sampling frequency of sensors [13], [14]. As a result, these methods may fail to deal with incomplete imaging modalities and suffer severe performance degradation. Consequently, it is of great significance to improve the robustness of multimodal remote sensing models for incomplete input.

To date, enormous efforts have been made to deal with multimodal remote sensing image classification with incomplete multimodal data. Generally, they can be divided into two types: generation-based methods and distillation-based methods. The generation-based methods synthesize the missing modalities by generative models and then apply the traditional multimodal algorithm directly [15], [16], [17], [18]. While this type of method is straightforward, the sample reconstruction may introduce detrimental noise and harm the model due to its complexity [19], [20]. Therefore, distillation-based methods are proposed to simplify the reconstruction problem from sample space to feature space via knowledge distillation [13], [14], [21], [22]. Despite the notable performance of these methods, they are customized. They assume that the missing modality is known and then train a model to recover it. This may be not applicable in real-world scenarios where the missing modalities are random and cannot be easily identified. Therefore, a generic model is highly desired for multimodal remote sensing classification to handle random and unknown missing modalities.

Common subspace-based learning has been widely adopted to tackle the issue of incomplete multimodal learning [23], [24], [25], [26], [27]. These methods remove modalities of each sample randomly during the training stage and guide the model to learn the invariant feature representation for all possible modality combinations, producing a generic deployment model capable of handling random input modalities. Specifically, early methods extract the invariant features by computing statistics, such as mean and variance [25]. To further improve the model performance, more recent methods design various attention models for information fusion to strengthen the representation ability of the invariant features [23], [24], [26], [27]. Despite the robustness of these methods, they may not achieve optimal performance since they ignore the specific information conveyed by each modality, a crucial factor in multimodal remote sensing classification [10].

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To address the above issues, we propose a general diversity-guided distillation network (DGDNet) for robust multimodal remote sensing image classification. Specifically, DGDNet consists of two sub-networks: the deployment network that extracts the modality-invariant representation for robust inference, and the teacher network that captures and transfers comprehensive multimodal information from complete multimodal data into the deployment network. With the joint supervision of vanilla classification loss and distillation loss, the deployment network can maintain the robustness to the random modality missing while capturing the invariant and specific information simultaneously for model optimization. In particular, we proposed a novel diversity-guided distillation (DGD) loss for DGDNet to train the deployment network by guiding it to mimic the feature diversity of the teacher network. Since multimodal feature diversity stems from inter-modality complementary information [28], [29], DGD can help the deployment model focus on learning the complementary modality-specific information, improving its representation ability. Moreover, since the feature diversity is semantically independent, DGD also overcomes the representational heterogeneity caused by input differences.

Besides, conventional multimodal learning usually faces the challenge of unbalanced training due to the variety of modalities. To be specific, the modality with better discrimination ability may acquire a larger gradient and dominate the optimization progress of the multimodal model [30]. As a result, the other modalities would suffer from under-optimized representation when the training is converged. This will not only limit the teacher network's ability to extract comprehensive information from multimodal data but also harm the robustness of deployment networks for hard modality combinations, especially when the modalities are highly heterogeneous. To cope with this problem, some methods are proposed to aid the training of multimodal models with the help of additional unimodal classifiers [23], [24] or pre-trained models [31]. While these methods improve performance to some degree, they require additional neural modules, complicating the training procedure.

To this end, we further propose a modality-center regularization (MCR) strategy to assist the training of DGDNet. Specifically, MCR alleviates the unbalanced training by constraining the distances of unimodal representation belonging to the same class across all modalities. This helps reduce the inter-modality discrepancy and thus can encourage the multimodal model network to pay fair attention to each modality. More importantly, this does not need the assistance of extra modules. Besides, MCR also considers minimizing the classification loss of each unimodal input. This helps improve the representation ability of modality encoders and enlarges the inter-class discrepancy. Here, MCR calculates the logit outputs via the fully-connected layer of the deployment network directly. Therefore, this also introduces no extra architectural complexity.

The major contributions of this work are summarized as follows.

1) We present a general framework DGDNet to assist robust multimodal remote sensing image classification by guiding the deployment network to learn comprehensive multimodal information from the teacher network trained with complete modality. This helps acquire the modality invariant and specific information simultaneously while maintaining robustness for incomplete modality input.
2) We propose the DGD strategy to transfer the comprehensive multimodal information to the deployment network by matching the feature discrepancy. This overcomes the representation heterogeneity when encouraging the deployment network to learn modality-specific information.
3) We propose the MCR strategy to minimize intra-class inter-modality distance and maximize intra-modality inter-class distance via parameter-shared constraint. This addresses the unbalanced training without extra architectural complexity.
4) Experiments and ablation studies demonstrate the effectiveness and generalization of the proposed framework for robust multimodal remote sensing image classification.

The rest of this article is organized as follows. Section II reviews related works, including the methods of multimodal remote seeing classification with missing modalities, common subspace-based learning, and unbalanced multimodal learning. Section III describes the details of the proposed DGDNet, including the overview framework, DGD, MCR, and training paradigm. Section IV provides the experimental results of three common multimodal remote sensing classification datasets from both a qualitative and quantitative perspective. Finally, Section V concludes this article.

II. Related Work

A. Multimodal Remote Sensing Classification With Complete Modalities

Recently, multiple remote sensing classification models have gained growing attention from the research community due to their great power in practice [32], [33], [34]. They can be divided into two categories: homogeneous-based methods and heterogeneous-based methods. Homogeneous-based methods process the modality data with similar physical properties, such as the fusion of hyperspectral (HS) and MS images [35], [36], as well as MS and PAN images [14]. Heterogeneous-based methods process the modality data with different physical properties, such as HS and light detection and ranging (LiDAR) images [10], HS and synthetic aperture radar (SAR) images [37], SAR and optical images [38], as well as LiDAR and optical images [39]. And many fusion methods have been proposed to handle multimodal remote sensing classification, such as attention-based fusion [7], [40] and common subspace-based fusion [10]. However, these methods usually assume that the models are trained and tested with the same modality data. This is not always possible in real-world deployment scenarios due to sensor failure [11], data corruption [12], and mismatched sampling frequency of sensors [13], [14].

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B. Multimodal Remote Sensing Classification With Missing Modalities

To tackle the challenge of missing modalities, researchers have proposed various approaches, which can be broadly categorized into two types: generation-based and distillation-based methods.

Generation-based methods synthesize the missing modalities by generative models and then apply the traditional multimodal algorithm directly. For example, Tran et al. [18] leverage the cascaded residual autoencoder to impute missing modalities. Recent works further explore the potential of the time-series information to resolve the reconstruction ambiguities [41], [42]. While these methods alleviate performance degradation, they are usually unstable and may introduce noise to harm the primary task at hand due to the complexity of sample reconstruction [19], [20].

Distillation-based methods are proposed to train a hallucination model to reconstruct the sample representation of the missing modality via knowledge distillation [43] and fuse it with the model trained with the available modality. This simplifies the reconstruction issue from sample space to feature space. Specifically, Kampffmeyer et al. [21] reconstruct the digital surface model (DSM) image representation to aid optical land cover classification with RGB images. Kumar et al. [14] reconstruct the representation of missing spectral data for multispectral land cover classification. Li et al. [22] introduce the dynamic-hierarchical attention distillation to improve the learning ability of the hallucination model. Wei et al. [44] propose a modality-shared hallucination network with joint adaptation distillation to preserve the cross-modal interaction for the hallucination model and improve its scalability for multiple missing modalities. Although these methods achieve promising results, the training of hallucination models requires the prior knowledge of missing modalities. This is difficult or infeasible in real-world scenarios where the modalities miss randomly.

C. Common Subspace-Based Learning

Common subspace-based learning methods aim to train one unified model to deal with different incomplete modality combinations by extracting the modality-invariant features. For example, HeMIS [25] learns an embedding of multimodal information by computing statistics (i.e., mean and variance) from any number of available modalities. Furthermore, Chen et al. [45] introduce the feature disentanglement to cancel out the modality-specific information. Besides, more recent methods, such as LCR [27] and RFNet [24], focus on extracting the modality-invariant representation via different attention mechanisms. Moreover, mmFormer [23] introduces the transformer block to model the global semantic information for the modality-invariant embedding. Wang et al. [46] propose the ShaSpec framework to capture the multimodal information with adversarial learning. While these methods achieve promising results, they only consider the modality-invariant information while ignoring the modality-specific information. As a result, they usually perform much worse than the customized methods, especially when more than one modality is missing.

D. Unbalanced Multimodal Learning

Due to the modality discrepancy, conventional multimodal learning models tend to fit the discriminative modality and their performance will degrade significantly when facing hard modality. To solve this issue, some studies introduce an auxiliary regularizer to perform classification tasks for each modality individually [23], [24], [47]. In this manner, each modality encoder is forced to learn discriminative features. Besides, there are also some works that propose to distill knowledge from the well-trained unimodal models to strengthen the unimodal encoders in multimodal learning [31]. These methods can indeed bring improvement to a degree, but extra efforts are required to introduce additional neural modules, which complicates the training procedure.

III. Method

A. Framework Overview

We propose a general framework DGDNet to assist the robust multimodal remote sensing classification with random missing modalities. Specifically, as shown in Fig. 1, DGDNet first introduces a "deployment network" to handle the multimodal remote sensing image classification with random missing modalities via modality dropout. For a specific, DGDNet introduces the Bernoulli indicators $\Delta = \{\delta^1, \ldots, \delta^m\}$ after modality encoders in the deployment network and randomly set some of them as 0 to simulate the real-world modality missing during the training stage. This encourages the deployment network to learn the invariant feature for all possible modality combinations, enabling it to deal with random modality missing during the inference stage. Then DGDNet introduces the "teacher network" to extract comprehensive multimodal information from complete multimodal data and transfer it to the deployment network via DGD. This enables the deployment network to acquire the modality-specific information. Finally, to deal with the unbalanced training problem, DGDNet further introduced the MCR module to assist the training of deployment and teacher networks by minimizing the inter-modality discrepancy and maximizing intra-modality inter-class distance.

In summary, we will first introduce the notations used in our methods. Then, we present the DGD and MCR in turn. Finally, we discuss the network training and testing details.

B. Notations

The notations used in this article are described as follows. $\Theta = \{x_i, y_i\}_{i=1}^N$ is a multimodal dataset that has N samples. Each x_i consists of V inputs from different modalities as $x_i = (x_i^1, \ldots, x_i^V)$. E_t^i and E_s^i denote the modality encoder for ith modality input in the teacher network and deployment networks, respectively. X_t^i and X_s^i denote the modality representation for ith modality input in teacher and deployment networks, respectively. $\Delta = \{\delta^{(1)}, \ldots, \delta^{(V)}\}$ is the Bernoulli sequence corresponding to the V modality encoders in the deployment network, where $\delta^i \in \{0, 1\}$. f_t and f_s denote the modality fusion module for the teacher and deployment network, respectively. $\tilde{X}_t \in R^{B\times C\times H\times W}$ and $\tilde{X}_s \in R^{B\times C\times H\times W}$ denote the fused representation in teacher and deployment

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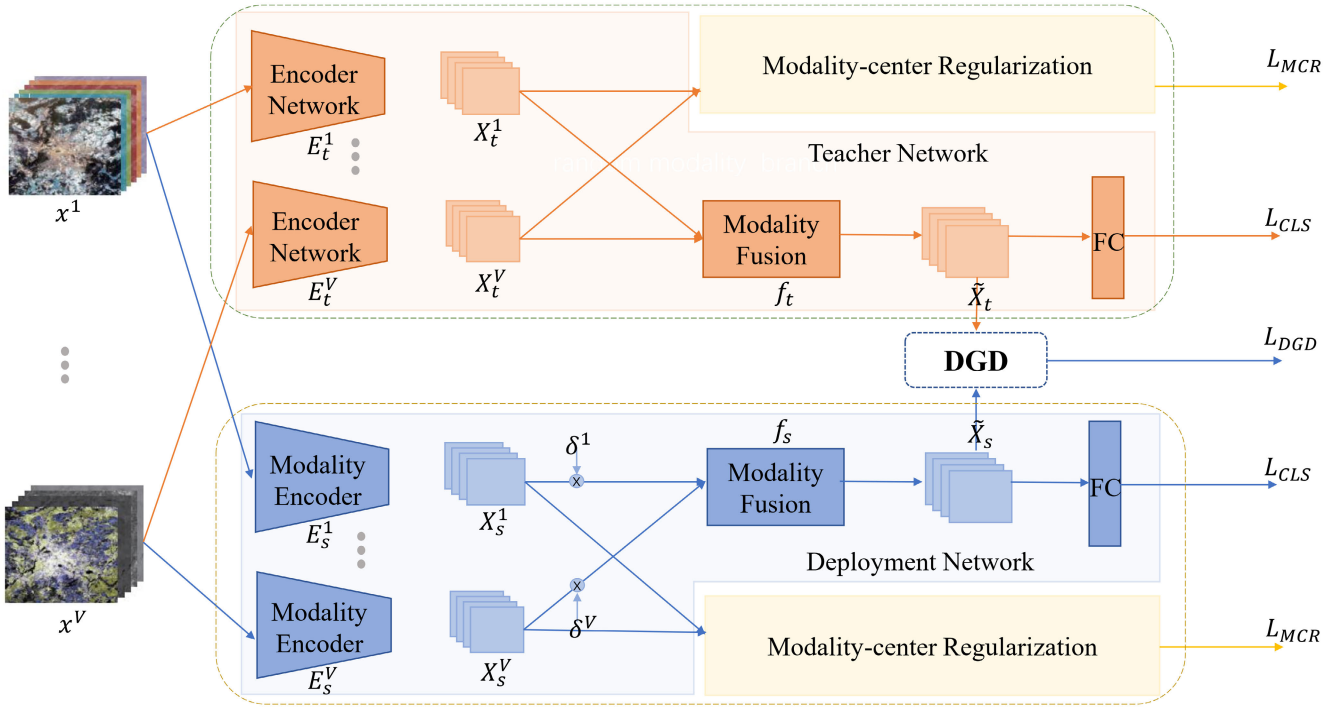


Fig. 1. Overall architecture of the proposed DGDNet. It consists of two parts: the deployment network that learns modality invariant features for final inference via the modality dropout, and the teacher network that captures and transfers comprehensive multimodal knowledge to the deployment network via the DGD method. Besides, DGDNet introduces the MCR for teacher and deployment networks to alleviate the unbalanced training. "FC" means fully-connected layer. Notations and training procedures are described in Sections III-B and III-E, respectively.

networks, respectively. Here, B, C, H, and W mean the batch size channel number, height, and width, respectively.

C. Diversity-Guided Distillation

Different from conventional distillation tasks where teacher and student networks have the same inputs, in DGDNet, $\tilde{X}_t$ and $\tilde{X}_s$ could be calculated from different modality combinations due to the modality dropout in deployment network. As a result, the conventional distillation methods that rely on feature consistency may fail in DGDNet and lead to sub-optimal performance due to input heterogeneity. To address this challenge, this section introduces the DGD strategy to transfer the comprehensive multimodal information from the teacher network to the deployment network by matching their feature diversity. On the one hand, the feature diversity is semantic-independent, so it will not be affected by input heterogeneity. On the other hand, feature diversity stems from inter-modality complementary information [28], [29], so DGD can encourage the deployment network to learn the modality-specific information, improving its representation ability.

As shown in Fig. 1, given a complete multimodal input $x_i = (x_i^1, \ldots, x_i^V)$, the teacher network will extract the fusion feature $\tilde{X}_t$ as follows:

$$\begin{cases} \tilde{X}_t = f_t\left(X_t^1, \ldots, X_{X_t}^V\right) & (1) \\ X_t^i = E_t^i(x_i). & (2) \end{cases}$$

To make the model robust for incomplete inference data, we leverage the modality dropout technique to generate the incomplete data for training. Specifically, we introduce the Bernoulli indicator $\delta \in [0, 1]$ after modality encoders and randomly set it to either 0 or 1, indicating whether the corresponding modality is missing or not. For missing modalities, the encoded feature maps will be replaced by a zero matrix. And the fused feature with modality dropout for the deployment can be rewritten as follows:

$$\begin{cases} \tilde{X}_s = f_s\left(X_t^1 * \delta^1, \ldots, X_{X_t}^V * \delta^V\right) & (3) \\ X_s^i = E_s^i(x_i). & (4) \end{cases}$$

Then, the DGD module is embedded into the penultimate layer of the teacher and deployment network. As described in Algorithm 1, it takes $\tilde{X}_t$ and $\tilde{X}_s$ as inputs and transfers the knowledge from the teacher and deployment network by minimizing the loss as follows:

$$L_{\text{DGD}} = \frac{1}{B^2} ||\mathcal{D}(\tilde{X}_t) - \mathcal{D}(\tilde{X}_s)||_F^2 \quad (5)$$

where $||.||_F$ is the Frobenius norm. $\mathcal{D}(X)$ is the feature diversity descriptor for the representation matrix X.

Since feature diversity can be measured by inter-feature discrepancy [48], [49], we can implement $\mathcal{D}(X)$ as follows:

$$\mathcal{D}(X) = 1 - \mathcal{P}(X) \quad (6)$$

where $\mathcal{P}(X)$ is the correlation matrix (i.e., the matrix of correlation coefficients) for X, whose elements range in $[0, 1]$. Specifically, we take the B sample representation vectors in a mini-batch as b random variables and calculate their correlation coefficients as the measure of feature similarity in this mini-batch. By subtracting the similarity, we can get the discrepancy of different sample representations as the measure

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Algorithm 1 Diversity-Guided Distillation

Input: The fused representation $\tilde{X}_t$ and $\tilde{X}_s$ from teacher and deployment networks.

Output: L_{DGD}

Covert the shape of $\tilde{X}_t$ and $\tilde{X}_s$ to $R^{B\times C*H*W}$

Calculate zero-centering matrix $\mathcal{C}(\tilde{X}_t)$ and $\mathcal{C}(\tilde{X}_s)$ via Eq. 8

Calculate correlation matrix $\mathcal{P}(\tilde{X}_t)$ and $\mathcal{P}(\tilde{X}_s)$ via Eq. 7

Measure feature diversity $\mathcal{D}(\tilde{X}_t)$ and $\mathcal{D}(\tilde{X}_s)$ via Eq. 6

Calculate diversity-guided distillation loss L_{DGD} via Eq. 5

Algorithm 2 Modality-Center Regularization

Input: The unimodal representation $X^1, \ldots, X^V$, transformer function $T(.)$ and fully-connect layer matrix W.

Output: L_{MCR}

Covert the unimodal representation into the feature vector $T(X^1), \ldots, T(X^V)$ for the fully-connect layer.

Calculate the logit output of each feature vector $W * T(X^1), \ldots, W * T(X^V)$.

Calculate L_{MCR} via Eq. 12

of feature diversity in this mini-batch. Compared with the previous methods that measure feature diversity via singular values [49], this avoids the computational complexity brought by the singular value decomposition.

To calculate the correlation matrix, we need to convert X from $R^{B\times C\times H\times W}$ to $R^{B\times C*H*W}$. Its rows represent different samples, and columns represent different features. Then, according to the definition of the correlation matrix, we can get $\mathcal{P}(X)$ as follows:

$$\mathcal{P}(X) = \frac{X_c * X_c^T}{\Sigma(X_c) * \Sigma(X_c)^T} \tag{7}$$

where $X_c = \mathcal{C}(X)$ is the zero-centering X. $\Sigma(X_c)$ is the variance matrix that consists of the variance of each row of X_c. Particularly, the zero-centering function $\mathcal{C}(X)$ is defined as follows:

$$\mathcal{C}(X_i) = X_i - \frac{\sum_{j=0}^{C*H*W-1} X[i,j]}{C*H*W} \tag{8}$$

where X_i means the ith row of X. The notation $[i, j]$ denotes the element at the ith row and jth column in X. This function performs zero-centering processing on sample representation in each row of X by subtracting its mean value.

D. Modality-Center Regularization

For DGD to work, the teacher model needs to be able to effectively extract comprehensive multimodal information. However, this is not necessarily satisfied due to unbalanced training. More specifically, conventional multimodal models tend to fit discriminative modalities while ignoring weak modalities [24], [31]. To tackle this issue, this section introduces MCR to alleviate unbalanced training between different modality encoders. Specifically, MCR proposes to minimize inter-modality discrepancy by constraining the distances of unimodal representation belonging to the same class across all modalities. This encourages the teacher network to pay fair attention to each modality and avoid the optimization bias for the weak ones. Moreover, it can also be applied to the deployment network to alleviate its unbalanced training problem.

Algorithm 2 shows the overall details of MCR. It regularizes the model training by minimizing the modality-center loss as follows:

$$L_{\text{MCR}} = \frac{1}{B}\sum_{i=1}^{B}\sum_{m=1}^{V}\left\|X_i^m - C_{y_i}\right\|_F^2 \tag{9}$$

where X_i^m is the modality representation of mth modality of the ith sample, y_i is the class label of ith sample, and C_{y_i} is the center representation of y_ith class. Different from the conventional hetero-center loss that finds a center for each modality and each class [26], MCR considers the relation among different modalities and thus can be more effective for reducing the intra-class variation.

To optimize L_{MCR}, we need to estimate the center representation of each class. Existing center-loss methods [26], [50], [51] treat C_{y_i} as a learnable parameter and update it in each iteration by averaging the features of the same class in the mini-batch. Despite its effectiveness, it introduces additional optimization parameters that may affect the learning of the task at hand.

To this end, we make two innovations.

1) According to the definition of the cross-entropy loss, for a well-trained CNN, its yith column of the full-connect layer matrix W_{y_i} share the similar directions of C_{y_i} [52], [53]. Therefore, we can approximate the C_{y_i} with W_{y_i} and rewrite L_{MCR} as follows:

$$L_{\text{MCR}} = \frac{1}{B}\sum_{i=1}^{B}\sum_{m=1}^{V}\left\|T\left(X_i^m\right) - W_{y_i}\right\|_F^2 \tag{10}$$

where $T(.)$ is the mapping function that matches the feature dimension of X_i and W_{y_i} by transferring the unimodal representation vector into the final feature vector for the full-connect layer. Here, the full-connect layer matrix is exactly that of the network using MCR. Besides $T(.)$ is the modality fusion function of the network using MCR, such as f_t in the teacher network and f_s in the deployment network.

Furthermore, since W_{y_i} is the y_ith column of the fully-connected layer matrix, we can get the equivalent constraint of (6) as follows:

$$L_{\text{MCR}} = \frac{1}{B}\sum_{i=1}^{B}\sum_{m=1}^{V}L_{\text{CLS}}\left(W * T\left(X_i^m\right), y_i\right) \tag{11}$$

where L_{CLS} is the cross-entropy loss.

2) Since the parameters of $T(.)$ and W are shared with those of the villain multimodal network, they will also be influenced by the classification loss of the fused feature. We further introduce the mean square error loss to constrain the logit output distance of different modalities. This helps further reduce the cross-modality

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discrepancy. And L_{MCR} can be rewritten as follows:

$$L_{\mathrm{MCR}} = \frac{1}{B}\sum_{i=1}^{B}\sum_{m=1}^{V} L_{\mathrm{CLS}}\left(W * T\left(X_i^m\right), y_i\right) + \frac{1}{B}\sum_{i=1}^{B}\sum_{m=1}^{V-1} L_{\mathrm{MSE}}\left(W * T\left(X_i^m\right), W * T\left(X^{m+1}\right)\right). \tag{12}$$

E. Training Paradigm

The training of the proposed DGDNet consists of two steps: teacher network training and deployment network training.

1) Teacher Network Training: The teacher network training aims to extract comprehensive multimodal information from the complete modality data for deployment network training. Therefore, the modality encoder module, fusion module, and fusion strategy of the teacher network are set as those in the deployment network. This helps reduce the structural difference, improving the transfer performance. The details can be seen in the deployment network training. The training loss L_t for teacher network is defined as follows:

$$L_t = L_{\mathrm{CLS}}(W(f_t(X^1, \ldots, X^V)), y) + \alpha L_{\mathrm{MCR}}(X^1, \ldots, X^V) \tag{13}$$

where the L_{CLS} term guides the teacher network to learn discriminative representation and the L_{MCR} term helps balance the training of different modality encoders to avoid the underfitting for weak modality. α is the balanced factor for them.

2) Deployment Network Training: The deployment network training aims to train a model that can acquire the modality invariant and specific information simultaneously while maintaining robustness for random missing modalities. Specifically, the architecture of the modality encoder is set as the MDL-CNNs-Net [54], which has been proven to be effective in multimodal land cover classification. It is worth noting that the DGDNet is model-agnostic and the multimodal encoder can also be implemented with other architectures, such as ResNet [5] and AlexNet [40]. The feature fusion module is implemented with the Fu-CNNs-Net along with MDL-CNNs-Net [54]. Also, the fusion module can also be implemented with other fusion strategies such as late fusion, cross fusion, and middle fusion.

The training loss L_s for the deployment network is defined as follows:

$$L_s = L_{\mathrm{CLS}}\left(W\left(f_t\left(\delta^1 X^1, \ldots, \delta^V X^V\right)\right), y\right) + \beta L_{\mathrm{DGD}}\left(X_s^f, \ldots, X_t^f\right) + \gamma L_{\mathrm{MCR}}\left(X^1, \ldots, X^V\right). \tag{14}$$

Different from the teacher network, the L_{CLS} term in deployment network training further takes the modality dropout into consideration. In detail, as shown in Fig. 1, each unimodal representation $\{X^i\}_{i=1}^V$ is multiplied by a Bernoulli indicator $\{\delta^i\}_{i=1}^V$ before feeding into the fusion module. The Bernoulli indicator will be randomly set as 0 during the training stage to simulate the random modality missing. This helps the deployment network to learn robust and discriminative features for all possible input modality combinations. Besides, L_{DGD} guides the deployment network to learn comprehensive multimodal information from the teacher network, enabling the deployment network to acquire modality-specific information. L_{MCR} term helps balance the training of different modality encoders to avoid underfitting for weak modality. β and γ are the balanced factors for them.

Algorithm 3 Training Paradigm of DGDNet

Training Step 1:
1) Train teacher network based on equation 13.
Training Step 2:
1) Initialize teacher network with weights learned in step 1.
2) Train deployment network based on equation 14.

IV. Experiments

A. Datasets

We evaluate the effectiveness of the proposed DGDNet on three widely used multimodal land cover classification datasets: HS-LiDAR Houston2013, HS-MS Houston2013, and Augsburg. A brief description of the three datasets is given as follows.

1) HS-LiDAR Houston2013: This dataset is a multimodal dataset consisting of HS and LiDAR images [54]. The HS image consists of 144 spectral bands ranging from 0.38 to 1.05 μm. The LiDAR image is a grayscale image of the same size as the HS image. In total, the dataset contains 15 ground object classes with 15 029 ground-truth samples, of which 2832 are used for training and 12 197 for testing.

2) HS-MS Houston2013: To make full use of the high spectral information of the HS image and high spatial information of the MS image, Hong et al. [10] release the HS-MS Houston2013 dataset from the HS data in HS-LiDAR Houston2013 by degrading the spatial and spectral domains. Specifically, the resulting MS image is composed of the same size as the original HS image and eight spectral bands at a ground sampling distance (GSD) of 2.5 m. In total, the dataset contains 15 ground object classes with 15 029 ground-truth samples, of which 2832 are used for training and 12 197 for testing.

3) Augsburg: This dataset includes three modalities, namely an HS image, a polarimetric SAR image, and a DSM image. The HS image comprises 180 bands covering the spectral range of 0.4–2.5 μm, while the SAR image consists of four features of the SAR covariance matrix. The DSM image is a single-band grayscale image, and the dimensions of each band are 332 × 485 pixels. In total, the dataset contains seven different classes and the ground-truth information is provided for 78 294 pixels, which are split into 761 for training and 77 533 for testing. Overall, this dataset provides a unique opportunity for the development and evaluation of algorithms for multimodal image fusion and classification.

B. Experiment Setup

1) Evaluation Metrics: In our study, we have selected the multimodal land cover classification task as a means of

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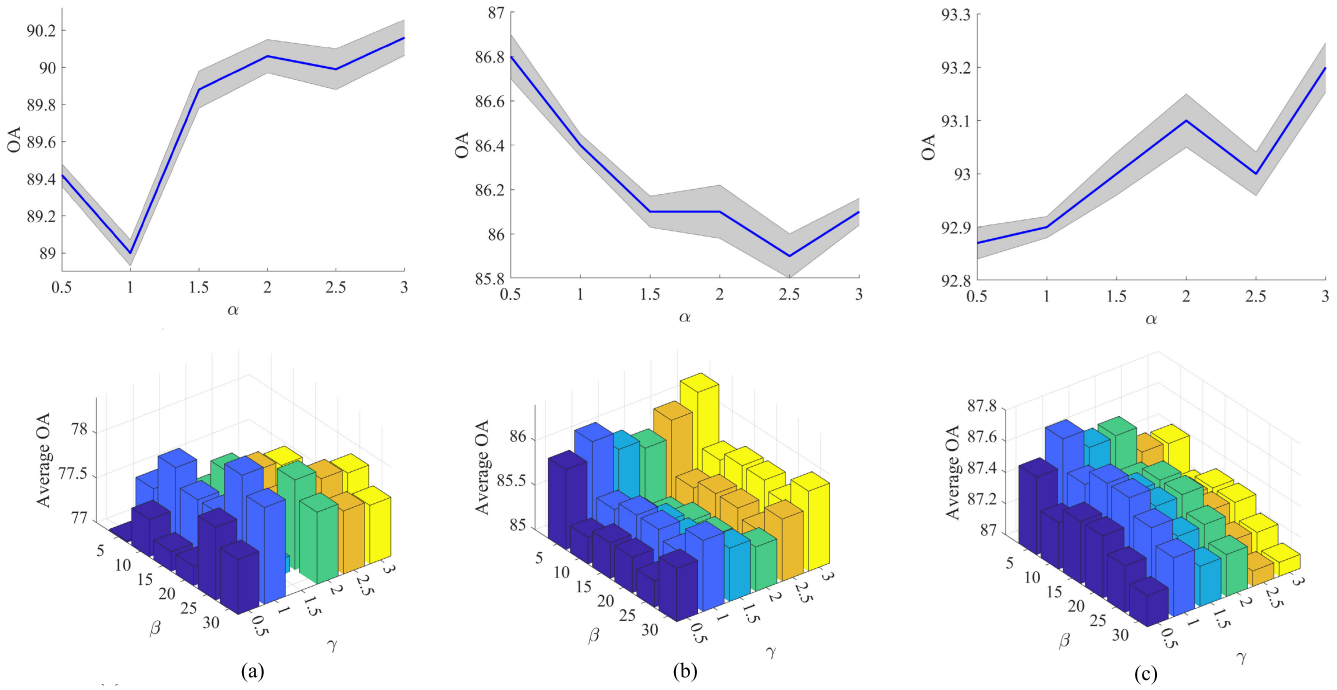


Fig. 2. Depiction of the trend between the OA and the hyperparameters α, β, and γ on three datasets. Specifically, the top line denotes the OA of the teacher model with different α on Huston2013 HS-LiDAR, Huston2013 HS-MS, and Augsburg datasets, respectively. The bottom line denotes the OA of the deployment model with different β and γ on (a) Huston2013 HS-LiDAR, (b) Huston2013 HS-MS, and (c) Augsburg datasets, respectively.

evaluating the effectiveness of the proposed framework. This task is considered to be a suitable application for assessing the performance of multimodal learning [10]. To provide a comprehensive assessment, we have employed three metrics from the field of land cover classification, namely the overall accuracy (OA), average accuracy (AA), and kappa coefficient (Ka).

2) Implementation Details: The PyTorch platform is utilized to implement the proposed method, which is trained using the stochastic gradient descent (SGD) optimizer with a training epoch of 500, batch size of 32, learning rate of 1e−3, and momentum of 0.9. Additionally, the classic cosine learning rate policy is employed to update the learning rate of each iteration for all models, which aids in achieving superior performance.

In addition to the training setting, data preparation plays a crucial role in the performance of the method. Unlike conventional classification tasks, where the entire image is used as the input to the network, remote sensing image classification requires cropping patches centered around a pixel as input. To ensure a fair comparison, we use the same patch cropping method as the previous work [54], where the original pixel is extended through the "replicate" operation, and the patch size is set to 7×7. Moreover, we employ random flipping as a data augmentation technique.

3) Comparison Methods: To assess the efficacy of DGDNet in addressing the challenge of land cover classification with randomly missing modalities, we conduct comparative experiments with five state-of-the-art methods for incomplete multimodal learning, including HeMIS [25], LCR [27], RFNet [24], mmFormer [23], and ShaSpec [46]. HeMIS and LCR are the early methods that focus on extracting modality-invariant features only. RFNet and mmFormer further improve the discrimination ability of unimodal input through the auxiliary regularizer. ShaSpec captures the multimodal information with adversarial learning. In general, these methods are uncustomized methods that train a generic model to handle random missing.

Besides, we introduce two baseline methods, customized MDL-CNN and uncustomized MDL-CNN. Specifically, MDL-CNN is a common multimodal baseline proposed in [54]. Customized MDL-CNN will train separate MDL-CNN models for each modality combination and measure its performance with the same modality. This provides the baseline of unimodal and multimodal learning with complete modality. The uncustomized MDL-CNN trained only one MDL-CNN model with the complete multimodal dataset, providing the baseline for incomplete multimodal learning.

Finally, for a fair comparison, we follow the basic implementation of MDL-CNN for all the comparison methods. Specifically, we unify the modality encoders and fusion strategy as the MDL-CNN-Net and parameter-shared fusion used in MDL-CNN. Besides, we set the MDL-CNN network trained with complete multimodal data as the teacher network and the MDL-CNN network with modality dropout as the deployment network.

C. Configuration of Hyperparameters

As discussed above, the proposed distillation strategies have three hyperparameters: α for the teacher network and β and γ for the deployment network, which would influence the land cover classification performance of our proposed method.

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Therefore, we conduct experiments on three datasets to search for the optimal hyperparameters. Specifically, the values of the parameters α and γ for MCR range from 0.5 to 3.0 with a span of 0.5. The values of parameter β for DGD range from 5 to 30 with a span of 5.

As shown in Fig. 2, the optimal α for HS-LiDAR Huston2013, HS-MS Huston2013, and Augsburg datasets are 3.0, 0.5, and 3.0 respectively. Here the optimal α of HS-LiDAR Huston2013 and Augsburg datasets is bigger. This is because the modalities in them are highly heterogeneous and require a bigger weight to balance the training. Besides, the optimal (β, γ) for HS-LiDAR Huston2013, HS-MS Huston2013, and Augsburg datasets are $(25, 1.0)$, $(5, 3.0)$, and $(5, 1.0)$, respectively.

D. Results on HS-LiDAR Houston2013 Dataset

This section conducts experiments on the HS-LiDAR Houston2013 dataset, to study its effectiveness on the highly heterogeneous modality combination. The results are shown in Table I.

From the result of the customized MDL-CNN that trains the separate MDL-CNN model for each modality combination individually, we can see that the model trained with HS and LiDAR modalities together outperforms that trained with either HS or LiDAR alone in all three metrics. This confirms the effectiveness of multimodal learning for remote sensing classification. Nevertheless, as shown in the column of uncustomized MDL-CNN, the performance of the multimodal model drops significantly when suffers incomplete input. Specifically, compared with the unimodal baseline, the OA decreases by 20.22% and 35.96% for HS and LiDAR input, respectively. This demonstrates the deficiency of conventional multimodal methods when suffering incomplete multimodal input. In contrast, the proposed DGDNet overcomes this issue. It achieves comparable or even better results than the customized MDL-CNN for all three modality combinations (HS, LiDAR, and HS + LiDAR) in three indices. Particularly, it outperforms the customized MDL-CNN by 1.06%, 1.5%, and 1.22% in average OA, AA, and Ka, respectively.

More significantly, the resulting deployment network trained by the DGDNet is capable of capturing the modality-specific information from the teacher model, yielding higher performance over the common subspace-based methods that only consider the modality-shared knowledge. Specifically, DGDNet achieves the best performance among uncustomized methods for all three modality combinations (HS, LiDAR, and HS + LiDAR) in three indices. Particularly, compared with the second-best unified method, i.e., ShaSpec, DGDNet improves the average OA, AA, and Ka by 0.76%, 1.35%, and 1.03%, respectively. This shows the superiority of our method on the incomplete multimodal classification task.

Finally, Fig. 1 shows the classification map of considered compared algorithms in the given scene. Taking the parking lot1 (in the top black box) and highway (in the bottom black box) as examples, the results of DGDNet are more complete and have fewer misclassified. In contrast, the classification maps of highways from the LCR, RFNet, mmFormer, and ShaSpec are intermittent. Besides, while the classification maps of the highway from HeMIS are complete, the classification maps parking lot1 from HeMIS are the most incomplete. These results further demonstrate the superiority of DGDNet.

E. Results on HS-MS Houston2013 Dataset

This section conducts experiments on the HS-MS Houston2013 dataset, to study its effectiveness on the homogeneous modality combination. The results are shown in Table II.

Although HS and MS are homogeneous and contain more shared information than HS and LiDAR, the performance of the uncustomized MDL-CNN also drops significantly when dealing with incomplete inputs. Specifically, compared with the unimodal baseline of customized MDL-CNN, the OA decreases by 11.75% and 25.29% for HS and MS input, respectively. In contrast, the proposed DGDNet achieves comparable or even better results than the customized MDL-CNN for all three modality combinations (HS, MS, and HS + MS) in three indices, demonstrating its effectiveness.

Besides, while the HS and MS are homogeneous and the complementary modality-specific information is less than that of HS and LiDAR, DGDNet also achieves the best performance for almost all three modality combinations (HS, LiDAR, and HS + HS + MS) in three indices except the AA in HS modality, and Ka in HeMIS. Moreover, for these two scenarios, DGDNet is also the second-best result. These results demonstrate that learning comprehensive multimodal information is also necessary and effective in the scene with the homogeneous modality.

F. Results on Augsburg Dataset

This section performs experiments on the Augsburg dataset to evaluate the scalability of the DGDNet in the scene with multiple missing modalities. The results are shown in Table III.

Compared with the baseline of customized MDL-CNN, the performance of uncustomized MDL-CNN also drops significantly when dealing with incomplete input data. More importantly, the drop increases with the number of missing modalities. Specifically, the OA decreases by 7.31%, 18.45%, and 7.01% when DSM, SAR, and HS modalities are missing, respectively. In contrast, the OA decreases by 25.34%, 18.47%, and 26.37% when DSM and SAR modalities, SAR and HS modalities, as well as HS and DSM modalities, are missing, respectively. These results reveal the importance of handling multiple missing modalities.

As shown in Table III, DGDNet outperforms all other uncustomized competitors in all metrics when multiple modalities are missing. Specifically, the OA of DGDNet outperforms that of the second-best method, i.e., ShaSpec, by 1.02%, 1.12%, and 1.14%, in average OA, AA, and Ka, respectively. Moreover, DGDNet also achieves the best performance for almost all modality combinations in three indices except the OA on SAR and AA metric on HS + SAR and HS + DSM. Even so, DGDNet is also the second-best result in them. Finally, DGDNet also achieves comparable or even better results than the customized MDL-CNN that trains models for all nine modality combinations in three indices. These results

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TABLE I

PERFORMANCE OF DIFFERENT METHODS ON THE HS-LIDAR HUSTON2013 DATASET. ● AND ○ DENOTE AVAILABLE AND MISSING MODALITIES, RESPECTIVELY. "CUSTOMIZED" MEANS TRAINING THE MODEL FOR EVERY MODALITY COMBINATION INDEPENDENTLY WHILE "UNCUSTOMIZED" MEANS TRAINING ONLY ONE MODEL FOR ALL MODALITY COMBINATIONS. THE BOLD AND UNDERLINE REPRESENT THE BEST AND SECOND-BEST RESULTS AMONG DIFFERENT UNCUSTOMIZED METHODS

Metric	Modality		Methods							
	HS	LiDAR	Customized	Uncustomized						
			MDL-CNN	MDL-CNN	HeMIS	LCR	RFNet	mmFormer	ShaSpec	DGDNet
OA	●	○	85.18	64.96	83.20	83.78	84.12	84.23	84.05	**84.58**
	○	●	58.53	22.57	50.56	53.67	56.74	58.29	58.92	**61.12**
	●	●	88.49	88.49	87.10	87.56	88.66	88.79	**89.33**	89.10
	Average		77.40	58.67	73.62	75.03	76.50	77.10	77.50	**78.26**
AA	●	○	86.60	86.12	74.78	86.31	85.43	85.58	85.92	**86.75**
	○	●	59.41	17.32	53.41	55.45	57.22	59.68	59.94	**62.68**
	●	●	89.54	89.54	88.20	87.93	89.10	89.42	89.92	**90.60**
	Average		78.51	59.03	75.92	76.56	77.25	78.22	78.66	**80.01**
Ka	●	○	83.89	62.13	81.65	82.65	82.79	82.49	82.82	**83.25**
	○	●	55.07	15.22	47.20	51.19	53.89	55.44	55.58	**58.15**
	●	●	87.50	87.50	85.74	85.88	87.47	87.81	88.61	**88.72**
	Average		75.48	54. 95	71.49	73.21	74.71	75.24	75.67	**76.70**

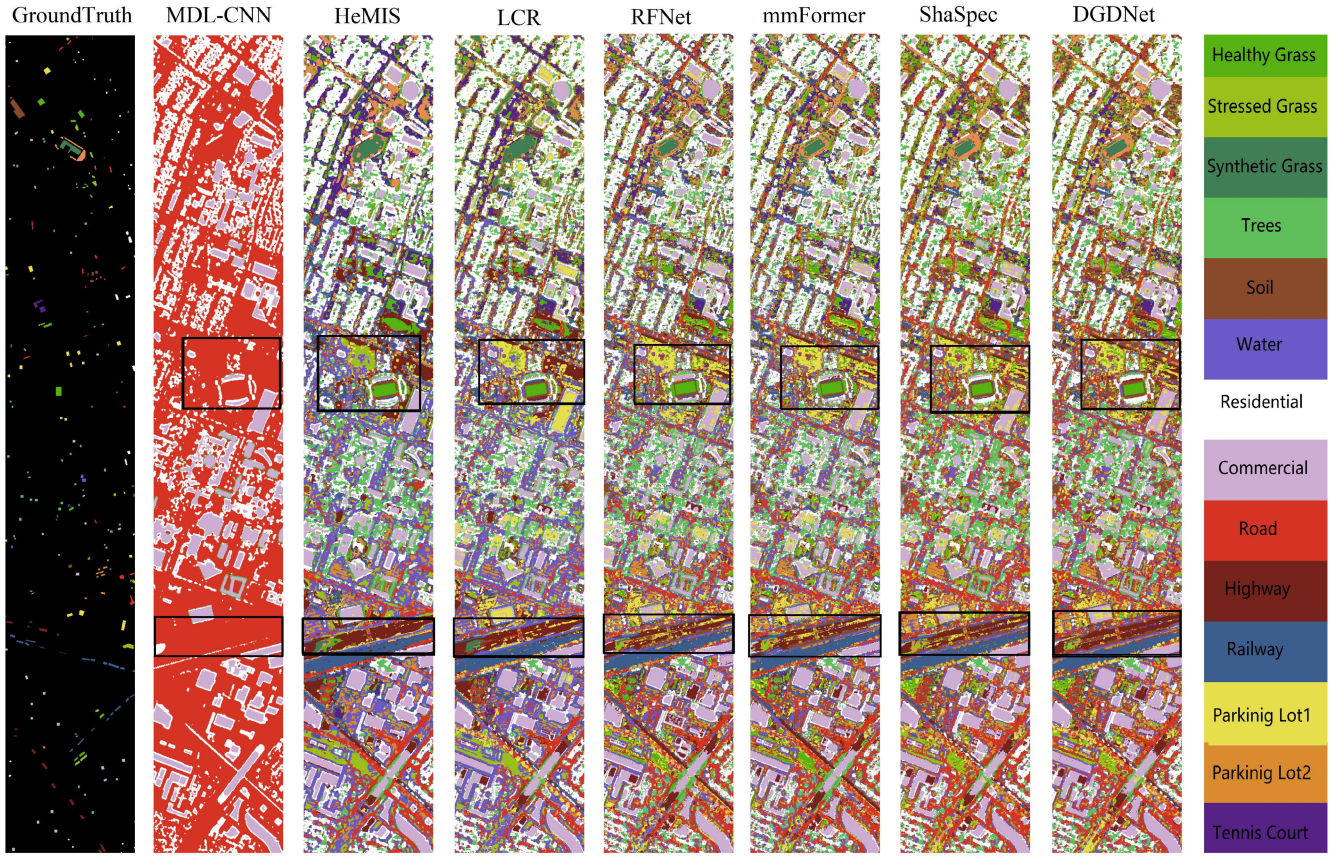


Fig. 3. Classification maps of different compared algorithms with LiDAR modality on Huston2013 HS-LiDAR dataset.

further confirm the superiority of DGDNet in dealing with multiple random missing modalities.

Finally, Fig. 4 shows the classification map of considered compared algorithms in the given scene. Take the forest (in the left black box) and residential area (in the right black box) as examples. The classification maps of the forest area from HeMIS are the most incomplete. Although mmFormer and ShaSpec also have a relatively complete classification map for the forest, they confused the forest and residential area. In contrast, the results of DGDNet on both areas are complete and have fewer misclassified.

G. Ablation Study

In this section, we first study the effectiveness of the important design elements in the proposed method, including the DGD and MCR. Then, we study the generalization of DGDNet

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TABLE II

PERFORMANCE OF DIFFERENT METHODS ON THE HS-MS HUSTON2013 DATASET. THE BOLD AND UNDERLINE REPRESENT THE BEST AND SECOND-BEST RESULTS AMONG DIFFERENT UNCUSTOMIZED METHODS

Metric	Modality		Methods							
	HS	MS	Customized	Uncustomized						
			MDL-CNN	MDL-CNN	HeMIS	LCR	RFNet	mmFormer	ShaSpec	DGDNet
OA	●	○	85.18	73.43	84. 48	84.02	84. 43	84.50	84.20	**85.44**
	○	●	86.11	60.82	84.45	83.65	84.57	84.8	84.31	**85.80**
	●	●	87.17	87.17	87.09	86.29	86.43	86.89	**87.67**	87.37
	Average		86.15	73.80	85.34	84.32	85.14	85.39	85.29	**86.27**
AA	●	○	86.60	76.33	86.43	85.94	86.39	**87.00**	86.44	86.88
	○	●	88.88	63.70	87.58	86.76	87.15	87.78	87.63	**88.33**
	●	●	90.01	90.01	89.81	87.08	88.46	89.55	**90.08**	89.87
	Average		88.49	76.68	88.27	86.59	87.33	88.11	88.05	**88.36**
Ka	●	○	83.89	69.21	84.16	82.73	83.01	83.47	82.17	**84.16**
	○	●	84.92	58.82	83.24	82.39	82.98	83.56	82.70	**84.12**
	●	●	86.51	86.51	86.21	83.67	84.78	85.71	**87.20**	86.78
	Average		85.10	71.51	84.73	82.93	83.59	84.24	84.20	**85.02**

TABLE III

PERFORMANCE OF DIFFERENT METHODS ON THE AUGSBURG DATASET. THE BOLD AND UNDERLINE REPRESENT THE BEST AND SECOND-BEST RESULTS AMONG DIFFERENT UNCUSTOMIZED METHODS

Metric	Modality			Methods							
	HS	SAR	DSM	Customized	Uncustomized						
				MDL-CNN	MDL-CNN	HeMIS	LCR	RFNet	mmFormer	ShaSpec	DGDNet
OA	●	○	○	90.91	65.65	89.19	89.61	89.76	89.99	90.61	**91.89**
	○	●	○	82.68	78.72	78.74	78.87	79.12	79.34	**80.11**	80.01
	○	○	●	78.12	51.75	69.59	73.47	74.89	75.58	76.03	**78.40**
	●	●	○	92.18	84.87	90.69	90.47	90.49	90.72	91.02	**91.55**
	●	○	●	90.49	72.04	90.70	90.68	90.66	90.89	91.11	**92.35**
	○	●	●	86.12	79.11	85.12	85.51	85.59	85.57	86.07	**86.55**
	●	●	●	93.31	91.39	91.69	91.47	91.42	91.33	91.68	**92.37**
	Average			87.68	72.71	85.10	85.72	85.99	86.14	86.56	**87.58**
AA	●	○	○	65.32	45.25	66.69	66.98	67.44	67.59	67.75	**69.34**
	○	●	○	46.12	24.87	39.51	40.12	42.79	43.81	43.37	**43.94**
	○	○	●	41.96	25.46	33.29	35.79	40.35	40.99	41.58	**42.36**
	●	●	○	63.74	62.73	**65.69**	65.14	64.57	64.78	65.33	65.55
	●	○	●	62.24	58.06	**67.97**	66.26	65.83	65.41	66.57	67.23
	○	●	●	45.23	36.80	48.11	48.01	48.41	48.54	48.11	**49.77**
	●	●	●	63.85	65.69	66.32	65.32	64.36	63.73	65.07	**65.42**
	Average			55.49	45.55	55.36	55.37	56.25	56.40	56.68	**57.80**
Ka	●	○	○	86.89	52.43	85.34	84.02	85.57	85.58	86.23	**87.43**
	○	●	○	74.75	43.59	68.32	63.65	69.42	69.73	70.63	**71.27**
	○	○	●	67.64	30.00	64.35	64.12	65.11	65.56	65.79	**68.02**
	●	●	○	88.59	76.48	85.30	85.09	85.41	85.97	86.48	**87.72**
	●	○	●	86.57	66.88	85.45	86.47	86.28	87.43	88.13	**88.96**
	○	●	●	79.33	69.47	79.26	78.29	78.73	79.72	79.62	**80.21**
	●	●	●	90.25	87.49	87.45	87.13	87.25	87.36	88.05	**89.29**
	Average			82.00	60.90	79.50	79.35	79.68	80.17	80.70	**81.84**

to different backbone networks. To this end, we conduct extensive ablation experiments on the modality combination of the HS-LiDAR Huston2013 dataset with different settings for the DGDNet.

1) Effect of DGD Strategy: To study the effect of the DGD strategy, we report and compare the performance of the DGDNet implemented with different distillation strategies. Specifically, the comparison distillation methods include the CMKD [55] that matches the value of feature maps, the RKD [56] that matches the distribution of logit output, and the JAD [57] that matches the distribution of feature maps. "N/A" means without the assistance of knowledge distillation.

As shown in Table IV, the model trained with knowledge distillation outperforms that without knowledge distillation in average OA, AA, and Ka, consistently. This demonstrates the effectiveness of learning comprehensive multimodal information from the teacher model. Moreover, the DGDNet implemented with the DGD strategy achieves the best results for all three modality combinations (HS, LiDAR, and HS + LiDAR) in three indices. Specifically, it outperforms the second-best result, JAD, by 0.57%, 0.71%, and 0.49% in average OA, AA, and Ka, respectively. These results confirm the effectiveness of transferring knowledge by matching the feature diversity.

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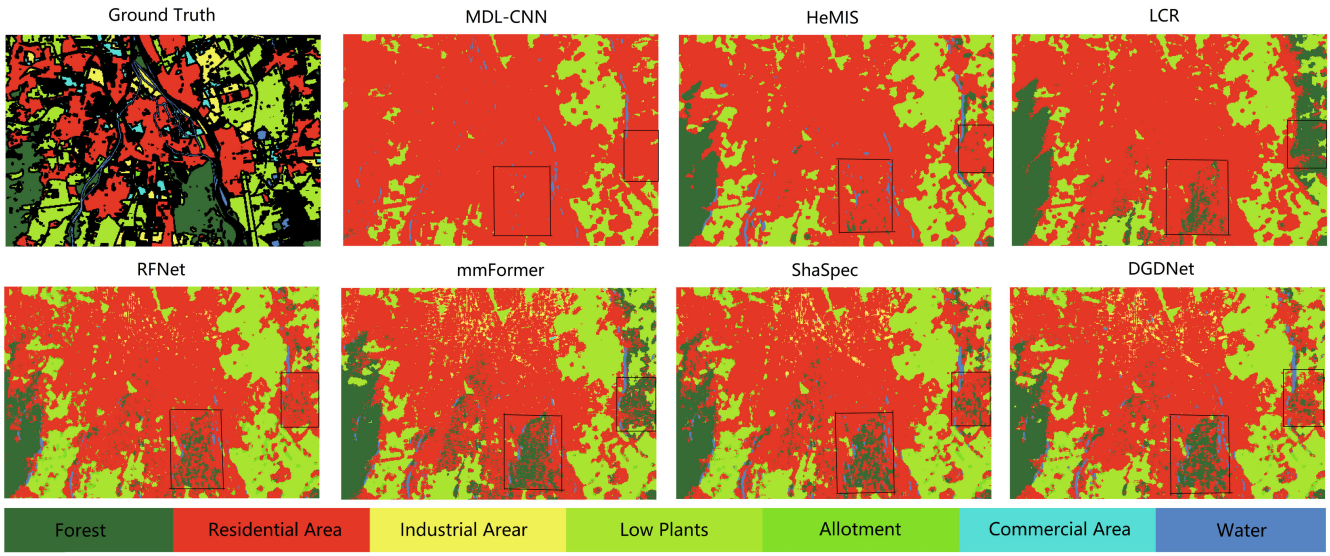


Fig. 4. Classification maps of different compared algorithms with DSM modality on Augsburg dataset.

TABLE IV
PERFORMANCE OF THE DGDNETS THAT ARE IMPLEMENTED WITH DIFFERENT DISTILLATION STRATEGIES ON THE HS-LIDAR HUSTON2013 DATASET WITH HS AND LIDAR MODALITIES

Metric	Modality		Methods				
	HS	LiDAR	N/A	CMKD	RKD	JAD	DGD
OA	●	○	83.76	82.85	83.93	84.55	**84.58**
	○	●	52.97	57.36	59.35	60.21	**61.12**
	●	●	88.09	87.13	88.12	88.63	**89.10**
	Average		74.94	75.78	77.13	77.79	**78.26**
AA	●	○	86.05	85.32	86.17	86.38	**86.75**
	○	●	54.33	59.55	60.76	61.72	**62.68**
	●	●	88.67	88.64	88.83	89.27	**90.06**
	Average		76.35	77.83	78.58	79.12	**79.83**
Ka	●	○	82.63	81.58	82.58	82.89	**83.25**
	○	●	49.31	55.01	56.34	57.56	**58.15**
	●	●	87.39	86.02	87.48	88.19	**88.72**
	Average		73.11	74.20	75.46	76.21	**76.70**

Besides, we also visualize the feature diversity of encoders for HS and LiDAR modality in the deployment networks with and without DGD in Fig. 5. From the figure, we can see that DGD can significantly improve the feature diversity of different modality encoders in the deployed network, demonstrating its effectiveness in transferring modality-specific information.

2) Effect of MCR Strategy: Here, we report and compare the performance of the teacher and deployment networks with and without the MCR to study its effect to alleviate the unbalanced training. As shown in Table V, MCR significantly improves the performance of both the teacher and deployment models when dealing with incomplete modal inputs. Specifically, for the teacher model, MCR improves the OA by 10.82% and 24.17% when LiDAR and HS modality are missing, respectively. More importantly, the performance gap between HS and LiDAR modality decreases from 42.39% to 29.54% in OA. Besides, for the deployment network, MCR improves the OA by 1.02%, and 1.13% LiDAR and HS modality are missing,

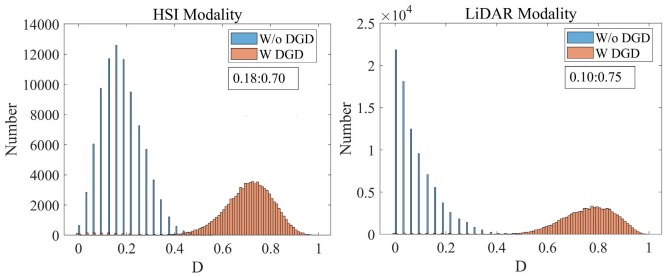


Fig. 5. Comparison of intra-modal feature diversity in the HS-LiDAR Huston2013 dataset. "W/" and "W/o" mean with and without the assistance of DGD. "D" denotes the inter-feature distance. "A:B" denotes the comparison of the average inter-feature distance of model "W/" and "W/o" DGD.

respectively. These results indicate that the multimodal model trained with MCR can extract more cross-modal information and has a more robust performance for modality missing. More specifically, MCR can avoid the model overfitting the discriminative modality and encourages it to pay attention to each modality fairly.

3) Effect of Patch Size: We report and compare the performance of ShaSpec and the proposed DGDNet implemented with different patch sizes. The results are shown in Fig. 6. As shown in the figure, the performance of ShaSpec and DGDNet will increase with patch sizes. This is because a bigger patch size will contain more information for a pixel, which leads to a better performance. Besides, for all patch sizes, the proposed DGDNet outperforms the second-best method, ShaSpec. This demonstrates the generalization of the DGDNet to different patch sizes.

4) Generalization to Different Backbone Networks: We report and compare the performance of the MD-CNN and DGDNet implemented with AlexNet [58] and ResNet [59] backbone to study the generalization of DGDNet to different backbone networks. Here, MD-CNN denotes the deployment network only using modality dropout. Both MD-CNN and DGDNet use the fusion parameter-shared strategy for

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TABLE V

PERFORMANCE OF THE TEACHER AND DEPLOYMENT NETWORK TRAINED WITH AND WITHOUT MCR ON THE HS-LIDAR HUSTON2013 DATASET

Metric	Modality		Method			
			Teacher Network		Deployment Network	
	HS	LiDAR	N/A	MCR	N/A	MCR
OA	●	○	64.96	**75.78**	84.06	**84.58**
	○	●	22.57	**46.74**	58.99	**61.12**
	●	●	**88.48**	88.39	**90.70**	89.10
	Average		58.67	**70.30**	77.93	**78.26**
AA	●	○	70.25	**78.74**	86.68	**86.75**
	○	●	17.32	**45.41**	61.03	**62.68**
	●	●	**89.54**	89.21	**90.76**	90.06
	Average		59.03	**71.12**	79.49	**79.83**
Ka	●	○	62.13	**73.87**	82.89	**83.25**
	○	●	15.22	**42.32**	57.56	**58.15**
	●	●	**87.50**	87.20	**88.96**	88.72
	Average		54.95	**67.79**	76.47	**76.70**

TABLE VI

PERFORMANCE OF THE MD-CNN AND DGDNET TRAINED WITH DIFFERENT BACKBONE NETWORKS. "MD-CNN" DENOTES THE DEPLOYMENT NETWORK USING MODALITY DROPOUT ONLY

Metric	Modality		Backbone			
			AlexNet		ResNet	
	HS	LiDAR	MD-CNN	DGDNet	MD-CNN	DGDNet
OA	●	○	79..60	**81.18**	81.67	**83.78**
	○	●	54.59	**64.66**	48.97	**54.76**
	●	●	84.28	**84.47**	85.03	**86.88**
	Average		69.43	**76.77**	71.89	**75.14**
AA	●	○	82.63	**83.59**	84.82	**85.38**
	○	●	56.12	**65.94**	51.66	**56.33**
	●	●	85.29	**85.58**	87.01	**88.57**
	Average		74.68	**78.37**	74.49	**76.76**
Ka	●	○	77.97	**79.69**	80.23	**81.71**
	○	●	51.03	**61.80**	45.34	**50.25**
	●	●	83.01	**83.17**	83.43	**84.95**
	Average		70.67	**74.88**	69.66	**72.30**

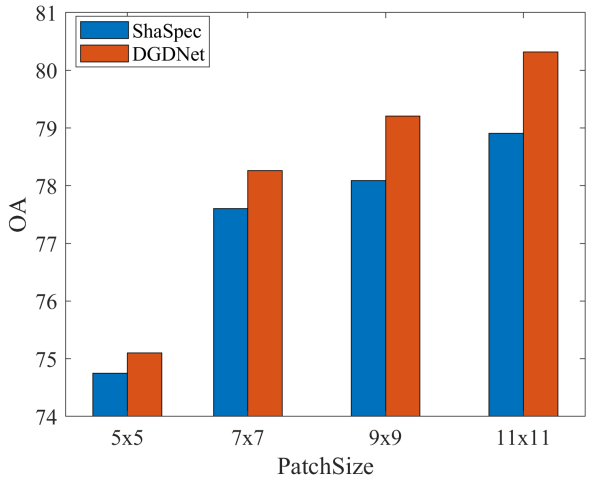


Fig. 6. Comparison of ShaSpec and DGDNet implement with different patch sizes.

information fusion. As shown in Table VI, DGDNet achieves the best performance for all modality combinations in all

TABLE VII

PERFORMANCE OF VANILLA DGDNET AND DGDNET* THAT COMBINES WITH SHASPEC METHODS ON HUSTON2013 HS-LIDAR DATASET

Metric	Modalities		ShaSpec	DGDNet	DGDNet*
	HS	LiDAR			
OA	●	○	84.05	84.58	**86.83**
	○	●	58.92	61.12	**61.27**
	●	●	89.33	89.10	**91.01**
	Average		77.50	78.26	**79.73**
AA	●	○	85.92	86.75	**88.37**
	○	●	59.94	62.68	**62.65**
	●	●	89.92	90.60	**91.90**
	Average		78.66	80.01	**80.97**
Ka	●	○	82.82	83.25	**85.73**
	○	●	55.58	58.15	**58.65**
	●	●	88.61	88.72	**90.11**
	Average		75.67	76.70	**78.16**

metrics for both Alexnet and ResNet networks. These results demonstrate the proposed DGD and MCR strategies can be generalized to different backbone networks.

5) Combination With SOTA Methods: Since the proposed DGDNet is independent of backbones and fusion methods, we further conduct experiments to study whether it can benefit from a better framework. To this end, we report and compare the performance of DGDNet implemented with MDL-CNN [54] (vanilla DGDNet) and the SOTA ShaSepc [46] (DGDNet*). As shown in Table VII, DGDNet* outperforms the vanilla DGDNet by 1.47%, 0.87%, and 1.46% on OA, AA, and Ka, respectively. These results demonstrate the scalability of DGDNet to the new framework and benefit from it.

V. CONCLUSION

In this article, we propose a general framework, called DGDNet, to assist the multimodal remote sensing image classification with random missing modalities. Specifically, DGDNet introduces a teacher network pre-trained with complete multimodal data to transfer the comprehensive multimodal information to the deployment network. This enables the deployment network to acquire the modality invariant and specific information simultaneously while maintaining robustness for incomplete modality input. In particular, a DGD module is embedded into DGDNet to conduct the knowledge transfer by matching the feature diversity. This helps overcome the representation heterogeneity while encouraging the deployment network to learn modality-specific information. Besides, we further introduce the MCR strategy to address the unbalanced training of teachers and deployment networks by constraining the intra-class inter-modality distance. This forces them to improve their representation ability for the weak modality. Finally, extensive experimental results on three datasets verify the effectiveness of our methods.

Limitations and future work: The proposed DGDNet requires the assistance of the teacher model to acquire the modality-specific information. On the one hand, this increases the training complexity. On the other hand, the training of the teacher model requires sufficient and complete multimodal data, hindering the application in few-shot task. Therefore,

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future work will investigate more efficient methods to capture the modality-specific information for robust multimodal remote-sensing classification.

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